This equation is known as *Brewster's law*. Brewster's law states that the tangent of the polarising angle for a transparent medium is equal to its refractive index. The polarising angle is known as Brewster's angle which dependes on the nature of the refracting medium.

EXAMPLE 7.17

Find the polarizing angles for (i) glass of refractive index 1.5 and (ii) water of refractive index 1.33.

Solution

Brewster's law, $\tan i_p = n$

For glass, $\tan i_p = 1.5;\ i_p = \tan^{-1} 1.5;\ i_p = 56.3^\circ$

For water, $\tan i_p = 1.33;\ i_p = \tan^{-1} 1.33;\ i_p = 53.1^\circ$

7.5.4.2 Pile of plates

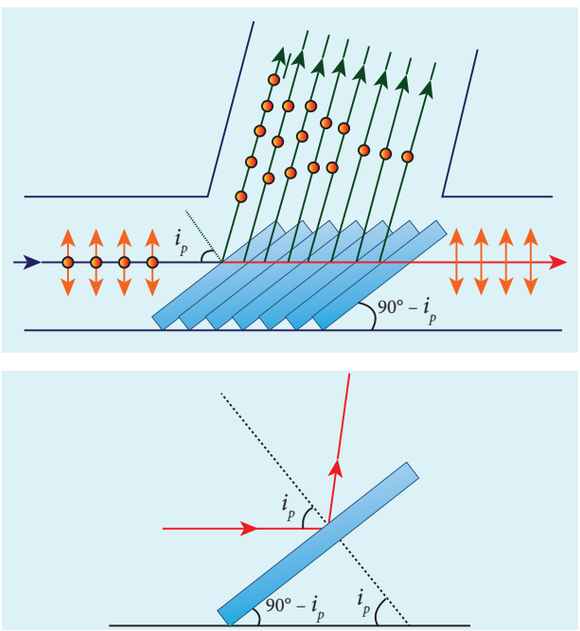


Figure 7.32 Pile of plates

Pile of plates makes use of Brewster's law to convert the partially polarised refracted light into plane polarised light. It consists of several glass plates kept one behind the other at an angle $90° - i_p$ with the horizontal surface as shown in Figure 7.32. This arrangement ensures that the parallel light falls on these plates at i_p. When this unpolarised light passes successively through these plates, the few parallel vibrations to the surface which may be present in the refracted light, get a chance for further reflections at the succeeding plates. Thus, both the reflected and the refracted lights are found to be plane polarised

EXAMPLE 7.18

What is the angle at which a glass plate of refractive index 1.65 is to be kept with respect to the horizontal surface so that an unpolarised light travelling horizontal after reflection from the glass plate is found to be plane polarised?

Solution

$n = 1.65$

Brewster's law, $\tan i_p = n$

$\tan i_p = 1.65;\ i_p = \tan^{-1} 1.65;\ i_p = 58.8^0$

The inclination with the horizontal surface is, $(90^0 - 58.8^0) = 31.2^0$

7.5.5 Polarisation by double refraction

Erasmus Bartholinus, a Danish Physicist discovered that **when a ray of unpolarised light is incident on a calcite crystal, two refracted rays are produced. Hence, two images of an object are formed. This phenomenon is called *double refraction* (or) *birefringence*** as shown in Figure 7.33.

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